

Impact of Kerr Nonlinearity within High Power Optical Amplifiers on HCF Transmission

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Abstract We quantify Kerr nonlinearity generated within C-band high-power EDFAs for long HCF links. We show that EDFA doped-fiber nonlinearity is the main contributor to system impairments, limiting allowable output power to below 32 dBm when the EDF length exceeds 10 m. ©2026 The Author(s)

Introduction

Recent advances in hollow core fiber (HCF) design and manufacturing have positioned HCF as a promising technology for next generation optical networks [1]. Due to its much higher tolerance to Kerr nonlinearity, significantly larger optical power can be launched into HCF without incurring nonlinear degradation [2]. This allows the use of higher-power optical amplification, which has been shown to increase system throughput in recent demonstrations [3-6]. However, prior studies have not examined the impact of Kerr nonlinearity originating within the high-power optical amplifiers (HPOAs) - specifically, nonlinearity generated in doped fibers such as Er-doped or Er/Yb-doped within HPOAs - leaving an important aspect of system performance unaddressed.

In this work, we analyse the impact of Kerr nonlinearity generated within C-band HPOAs on the throughput of a long, single-span HCF transmission link. We show that although Kerr effects in HCF are negligible, they are non-negligible in HPOAs, reducing both the allowable launch power and the throughput. Thus, for EDF lengths exceeding 10 m and with non-ideal transponders, operating HPOAs above 32 dBm may not be optimal for C-band networks.

Model and Parameters

One of the key advantages of anti-resonant hollow-core fibers (AR-HCFs) is their low attenuation. Recent reports show that AR-HCFs can achieve attenuation as low as 0.1 dB/km, or even

lower [1]. As a result, low loss and low Kerr nonlinearity HCF, in combination with HPOAs, can enable transmission over long spans (e.g., 160 km). This reduces the required number of in-line amplifiers, thereby lowering power consumption and system cost.

In this paper, we study the impact of Kerr nonlinearity originating in HPOAs and its contribution to the total nonlinear noise generated in HCF transmission links. Since HPOAs can be based on Er-doped or Er/Yb-doped fibers, for simplicity in the rest of the paper, we refer to HPOAs as erbium-doped fiber amplifiers (EDFAs) and to the doped fibers as erbium-doped fiber (EDF). We focus on the Kerr nonlinearity generated within the EDF sections and how it propagates through the overall EDFA-HCF-EDFA link (see Fig. 1 (a)). To assess the severity of Kerr nonlinearity in the EDFA, knowledge of the optical parameters of the EDF itself is required. However, the design of HPOAs is complex, and the corresponding EDF parameters are typically not publicly available. While some doped fibers designed for C-band EDFA have small effective area and large nonlinear index [7], others have large mode area for medium power, low nonlinearity applications [8], the exact parameters of EDFs used in high-power EDFAs remain unknown. In the absence of detailed specifications, we analyse Kerr nonlinearity using representative EDF parameters from [7] and investigate the impact of EDFA output power, EDF length and mode effective area.

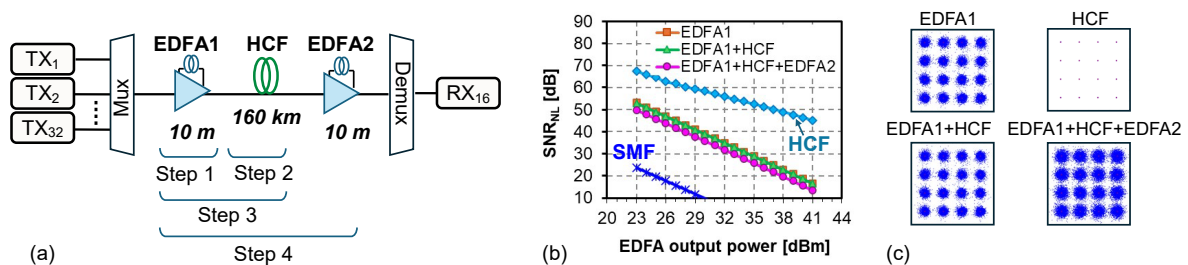


Fig. 1: (a) Schematic diagram of simulation studies; (b) SNR_{NL} vs EDFA output power; (c) constellation diagrams after transmission through step 1, 2, 3 and 4 at maximum output power of 41 dBm.

The simulation setup is shown in Fig. 1(a). The 10 m EDFs and 160 km HCF sections are concatenated to evaluate the nonlinear noise accumulation arising from both EDF nonlinearity and long span HCF transmission. The NLI noise is monitored after transmission through multiple steps: (step 1) the EDFA1 alone, (step 2) the HCF alone, (step 3) the combined EDFA1+HCF and (step 4) the combined EDFA1+HCF+EDFA2 configuration. Both pre- and post- EDFAs are assumed to have identical EDFs and power characteristics. The total amplifier output power varies from 23 dBm (typical) to 41 dBm (high power) in 1 dB increments. The EDFA is modelled under the no-pump-depletion assumption; accordingly, the signal power increases with the gain. The gain is set to compensate for the HCF span loss.

The transmitted signal consists of 32 channels of 130 GBaud DP-16QAM with a channel spacing of 150 GHz in the C band, resulting in a total occupied bandwidth of 4.8 THz. The channel under test (CUT) is the central channel. The transmission line comprises a single 160 km HCF span with an attenuation coefficient of 0.1 dB/km, corresponding to a total span loss of 16 dB, and splice losses are neglected. The impact of gas absorption lines within the signal spectrum is also neglected. The HCF and EDF parameters are set as follows: chromatic dispersion coefficients of 3.2 and 10 ps/nm/km, dispersion slopes of 4.1 and 57 ps/nm²/km, and nonlinear coefficients of 5×10^{-4} and $6.32 \text{ W}^{-1} \text{ km}^{-1}$, respectively, evaluated at a reference frequency of 193.75 THz.

Nonlinear noise limited signal-to-noise ratio (SNR_{NL}) contributions from both the EDF and the HCF are evaluated using split step Fourier method (SSFM) implemented in commercially available VPI DesignSuite software. Owing to the very short length of the EDF (on the order of meters) compared to the HCF span length (kilometers), the CF-GN model may not be applicable here, as it assumes Gaussian distributed NLI noise. It may give less accurate results in short, single span transmission links, which may exhibit non-Gaussian noise properties. To isolate nonlinear effects from ASE noise, noiseless amplification is assumed and transmitter and receiver DSP blocks are excluded from the simulation. Polarization mode dispersion is set to zero, and accumulated chromatic dispersion is fully compensated in the optical domain. The SNR_{NL} is estimated from the received constellation diagrams.

The amplified spontaneous emission (ASE) contribution, SNR_{ASE} , is calculated as:

$$\text{SNR}_{\text{ASE}} = P_{\text{in}} / NFhf_c B$$

where P_{in} is the input power to the EDFA, NF is the noise figure = 5 dB, h is Planck's constant, f_c

is the signal frequency, B is the optical signal bandwidth. The generalized SNR (GSNR) is obtained as:

$$\text{GSNR} = \left(\frac{1}{\text{SNR}_{\text{NL}}} + \frac{1}{\text{SNR}_{\text{ASE}}} + \frac{1}{\text{SNR}_{\text{TRX}}} \right)^{-1} \quad (1),$$

where SNR_{TRX} accounts for transponder electronic noise. Finally, the achievable channel throughput is estimated as:

$$C = 2B \log_2(1 + \text{GSNR}),$$

where B is the bandwidth of the CUT.

Results

Fig. 1(b) shows SNR_{NL} versus EDFA output power after transmission through four steps: 10 m EDFA1; 160 km HCF; 10 m EDFA1 + 160 km HCF; and 10 m EDFA1 + 160 km HCF + 10 m EDFA2. These results show that Kerr nonlinearity originating from the EDFA1 (orange line, almost overlapped with green line) is non-negligible. Compared with HCF, this contribution is considerably stronger and therefore represents the dominant source of signal degradation. Accordingly, the SNR_{NL} curve for the EDFA1+HCF (green line) closely overlaps with that of EDFA1 alone, and it degrades slightly more for the EDFA1+HCF+EDFA2 configuration. For comparison, the SNR_{NL} curve for 80 km SMF with loss coefficient of 0.2 dB/km (same 16 dB span loss as in HCF) is shown as blue line. The performance is approximately 29 dB worse than EDFA1, indicating that Kerr nonlinearity in the EDFA is negligible for SMF transmission links. The constellation diagrams at the highest EDFA output power $P_{\text{out}} = 41$ dBm are shown in Fig. 1(c). It shows that NLI noise generated by EDF at very high power of 41 dBm remains Gaussian-like due to strong contribution of four-wave-mixing (FWM). This observation highlights the impact of FWM in EDF due to short fiber length and high-power operating condition. The FWM crosstalk (XT) in EDFA can be suppressed by reducing EDF fiber length and/or increasing the effective mode area (A_{eff}).

Fig. 2 shows the channel throughput as function of EDFA output power for HCF-only and EDFA1+HCF+EDFA2 configurations, when the EDF length is set to 5, 10 and 30 m. Our goal is to investigate FWM crosstalk suppression for the range of EDF lengths, which are practical in C-band EDFAs. We see that for an ideal transponder (no SNR_{TRX} in Eq. (1)), the throughput exhibits an optimum that corresponds to the crossing point between SNR_{NLI} and SNR_{ASE} . Because EDF has lower tolerance to Kerr nonlinearity, SNR_{NLI} in EDFA1+HCF+EDFA2 is smaller, and the crossing point occurs at a lower optimum power than for HCF. Here, the highest throughput is achieved for EDF length of 5 m, where

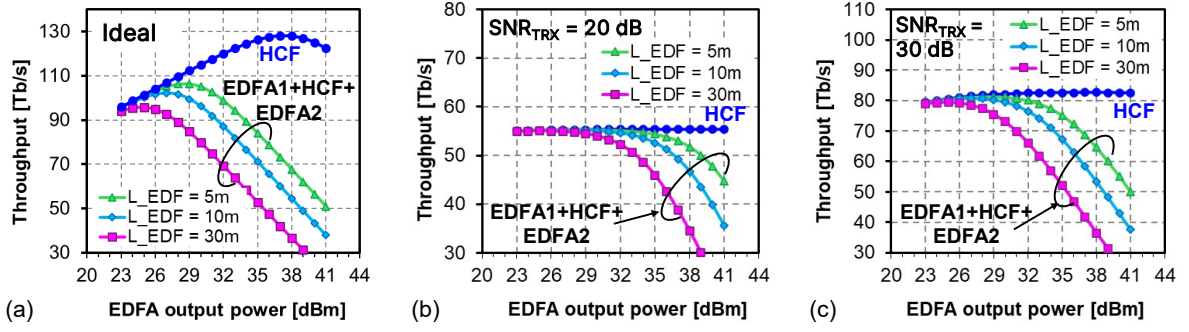


Fig. 2. Channel throughput and dependency on EDF length as a function of EDFA output power for: (a) ideal transponder; (b) non-ideal transponder with $\text{SNR}_{\text{TRX}} = 20$ dB and (c) with $\text{SNR}_{\text{TRX}} = 30$ dB.

suppression of FWM XT is more effective. But even with this short EDF length, NLI noise generated in EDF is still the dominant source of signal degradation. The optimum power decreases to 25 dBm and throughput reduces by 25% as the EDF length increases to 30 m.

For non-ideal transponders with $\text{SNR}_{\text{TRX}} = 20$ dB (typical) and $\text{SNR}_{\text{TRX}} = 30$ dB (high performance TRX) [9], the maximum throughput is further limited by SNR_{TRX} (Fig. 2(b) and 2(c)). We observe that there is an allowable range of EDFA output power, beyond which the achievable throughput starts to rapidly decrease. Increasing SNR_{TRX} from 20 to 30 dB raises the maximum achievable throughput by about 25 Tb/s, but it also tightens the allowable EDFA power range.

Next, we define the maximum allowable EDFA output power as the power beyond which the throughput degrades by more than 1% from the peak throughput. This maximum allowable EDFA output power can help with amplifier hardware design. Figure 3(a) (derived from Fig. 2(b) and Fig. 2(c)) shows that this maximum allowable EDFA power decreases significantly as the EDF length increases. For example, for $\text{SNR}_{\text{TRX}} = 20$ dB, the maximum allowable EDFA output power is 32 dBm for 10 m of EDF and 29 dBm for 30 m of EDF. These results indicate that very high-output in-line EDFAs are generally unnecessary;

operating above these values may degrade the throughput.

We next examine the effect of FWM suppression in a large effective mode area, low-nonlinearity EDFA. Fig. 3(b) shows that increasing A_{eff} reduces Kerr nonlinearity and, thus, shifts the maximum allowable EDFA power to higher values. Therefore, EDF length and amplifier design parameters should be selected carefully for HCF links to avoid throughput degradation.

Conclusions

We analysed the impact of Kerr nonlinearity generated in C-band HPOAs on the throughput of a 160 km single-span HCF transmission link. Although HCF exhibits high tolerance to Kerr nonlinearity, HPOAs may introduce stronger nonlinear impairments than the HCF span, thereby limiting the achievable throughput and reducing the maximum allowable amplifier output power. For non-ideal transponders, operating HPOAs above 32 dBm with EDF length longer than 10 m may not be optimal for HCF networks. These results highlight the importance of considering HPOA parameters and characteristics when designing HCF transmission links.

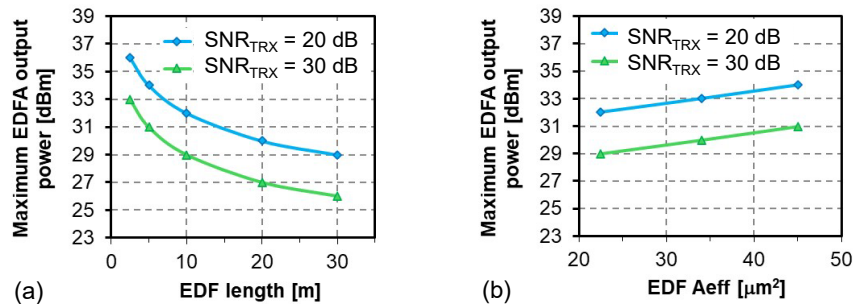


Fig. 3. The maximum allowable EDFA output power that still achieves peak throughput as a function of: (a) EDF length when $A_{\text{eff}} = 22.5 \mu\text{m}^2$; (b) EDF A_{eff} when the EDF length is 10 m.

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